

LESSON 5-2 • SESSION 1 • GUIDED NOTES

Area of Triangles

Learning target – I can find the area of a triangle using one half the base times the height. · 6.G.A.1

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
right triangle <i>triángulo rectángulo</i>	A triangle with one _____ corner.	its two legs are b and h
height <i>altura</i>	The distance from the base to the opposite corner, at a _____ angle.	h in $A = \frac{1}{2} \times b \times h$
congruent <i>congruente</i>	Exactly the same _____ and shape.	2 congruent triangles make a parallelogram

KEY IDEA*complete each sentence*

- Two congruent triangles make a _____.
- So a triangle's area is _____ of base $\times$ height: $A = \frac{1}{2} \times b \times h$.
- In a right triangle, the two _____ are the base and the height.

WORKED EXAMPLE*fill in the blanks as we solve*

A triangle has a base of 14 cm and a height of 9 cm. Find its area.

Step 1 $14 \times 9 = \underline{\hspace{2cm}}$ **Step 2** 126 cm^2 is the area of the parallelogram made by two triangles.**Step 3** $126 \div 2 = \underline{\hspace{2cm}}$ **Answer** The area is $\underline{\hspace{2cm}} \text{ cm}^2$.**YOU TRY***use the example as a model*

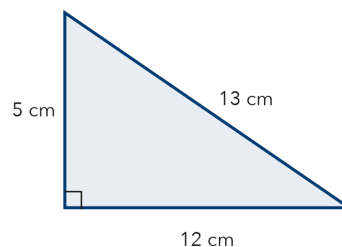
1. Find the area of each triangle.

a. $b = 10 \text{ m}, h = 7 \text{ m}$ _____ m^2

b. $b = 8 \text{ ft}, h = 5 \text{ ft}$ _____ ft^2

2. Find the area of the right triangle.

Show your work.

Answer _____ cm^2 **SUM IT UP***in your own words*The triangle formula has a $\frac{1}{2}$ because ____.

LESSON 5-2 • SESSION 2 • GUIDED NOTES**Missing Measures in Triangles**

Learning target – I can find a missing base or height of a triangle when I know its area. · 6.G.A.1

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
area <i>área</i>	The space inside a shape, measured in _____ units.	40 cm ²
formula <i>fórmula</i>	A rule written with _____ and symbols.	$A = \frac{1}{2} \times b \times h$
inverse operation <i>operación inversa</i>	An operation that _____ another.	doubling undoes taking half

KEY IDEA*complete each sentence*

- Undo the half first: _____ the area.
- Then _____ by the measure you know.
- A base or height is a length, so use units like cm, not _____.

WORKED EXAMPLE*fill in the blanks as we solve*A triangle has an area of 40 cm² and a base of 10 cm. Find the height.**Step 1** $40 = \frac{1}{2} \times 10 \times h$ **Step 2** Double the area: $2 \times 40 = \underline{\hspace{2cm}}$ **Step 3** Divide by the base: $80 \div 10 = \underline{\hspace{2cm}}$ **Answer** The height is 8 cm.**YOU TRY***use the example as a model*

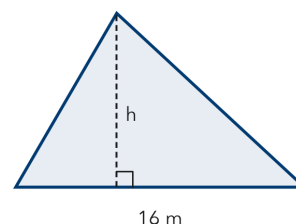
1. Find the missing measure.

a. $A = 30 \text{ ft}^2$, $h = 6 \text{ ft}$; $b = \underline{\hspace{2cm}}$ ft

b. $A = 21 \text{ m}^2$, $b = 7 \text{ m}$; $h = \underline{\hspace{2cm}}$ m

2. The area of the triangle is 48 m
- ²
- . Find the height
- h
- .

Show your work.

Answer _____ m**SUM IT UP***in your own words*

To find a missing base or height, I double the area first because ____.